

Economics 880 – PSet 2

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Part I

We solve for the competitive equilibrium in the setting of Hugget (1993) with enforceable insurance markets and no initial asset holdings. There are a unit measure of identical agents with preferences $\mathbb{E}_0[\sum_t \beta^t U(c_t)]$ where $U : \mathbb{R}_+ \rightarrow \mathbb{R}$ is strictly increasing, strictly concave, differentiable and bounded on its domain. As usual, $\beta \in (0, 1)$. The support of income shocks is $S = \{u, e\}$ where income is $y : S \rightarrow \mathbb{R}$ with $y(e) = 1$ and $y(u) = b < 1$. Shocks are iid across agents and follow a Markov process:

$$\pi(s' | s) = \Pr(s_{t+1} = s' | s_t = s).$$

Each period, agents' income is realized, and they choose asset holdings $a_{t+1} \geq \underline{a}$ given a price $q_t \in [0, 1]$ with consumption c_t determined by the residual income from asset holding decisions and where $\underline{a} \leq 0$.

We solve for the competitive equilibrium by first solving a planner's problem that weights each agent equally, and then decentralizing the planner's allocation via a bond price. While agents are subject to income risk, in aggregate the Law of Large Numbers implies no aggregate uncertainty. Thus, the planner's problem is given by

$$\max_{\{(c_t^e, c_t^u)\}} \sum_{t=0}^{\infty} \beta^t [\pi^e U(c_t^e) + \pi^u U(c_t^u)] \quad \text{s.t.} \quad \pi^e y(e) + \pi^u y(u) \leq \pi^e c_t^e + \pi^u c_t^u$$

where π^s is the fraction of the population that receives income shock s each period. With multiplier λ_t on the period t resource feasibility constraint, the first order optimality conditions of the associated Lagrangian are

$$\begin{aligned} \beta^t U'(c_t^s) - \lambda_t &= 0 \quad s = e, u \\ \lambda_t [\pi^e (c_t^e - y(e)) + \pi^u (c_t^u - y(u))] &= 0 \end{aligned}$$

Strict concavity of the utility function implies that $c_t^e = c_t^u = \pi^e y(e) + \pi^u y(u)$ for all t . In other words, there is perfect consumption smoothing between those agents suffering different income shocks.

The household problem is

$$\begin{aligned} \max_{\{(c_t, a_t)\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(c_t(s_t)) \quad \text{s.t.} \quad c_t + \sum_{s \in S} Q(s | s_t) a_{t+1}(s, s_t) &\leq y(s_t) + a_t(s_t, s_{t-1}) \\ a_0 &= 0 \end{aligned}$$

The consumption Euler equation for this problem is

$$Q(s' | s)U'(c_t(s)) = \beta\pi(s' | s)U'(c_{t+1}(s')).$$

To match the planner's allocation, the marginal utilities of consumption in adjacent periods must be equal as consumption is smoothed across time and states. Hence,

$$Q(s' | s) = \beta\pi(s' | s),$$

implying that the risk free rate is

$$q \equiv \sum_{s' \in S} Q(s' | s) = \beta.$$

Part II

Figure 1a plots the equilibrium policy functions for employed and unemployed households. For employed households, the policy function first satisfies $g(a, e) < a$ at approximately $a = 1.4404$. For unemployed households, the policy function first satisfies $g(a, u) < a$ at approximately $a = -1.9930$. Thus, for both earnings states, there exist asset levels $\bar{a}$ such that

$$g(\bar{a}, s) < \bar{a},$$

verifying that households do not continue to accumulate assets at the upper end of the asset grid.

The equilibrium bond price is $q^* = 0.9943$. At the reported equilibrium price, aggregate excess asset demand is approximately 0.00118, which is close to the zero-net-supply market-clearing condition. Because asset choices are restricted to a grid of 1,000 points, the numerical excess-demand function might only be roughly equal zero. We therefore terminate the bisection algorithm when the price bracket becomes sufficiently small rather than requiring excess asset demand itself to equal zero.

Figure 1b plots the cross-sectional wealth distributions for employed and unemployed households, where, wealth is defined as current earnings plus net asset holdings, $w = s + a$.

The stationary distribution assigns approximately 94.34% of households to the employed state and 5.66% to the unemployed state. Mean wealth in the stationary distribution is approximately 0.9729.

Figure 1c plots the Lorenz curve implied by the stationary wealth distribution. Because some households have negative net wealth, the Lorenz curve initially falls below zero before rising toward one. The resulting Gini coefficient is

$$G = 0.3854.$$

This is substantially below the wealth Gini of 0.80 observed in the data. Thus, despite generating endogenous wealth inequality through uninsurable idiosyncratic earnings risk, the benchmark incomplete-markets

model generates only about half of the wealth inequality observed in the data. Our estimate is also nearly identical to the benchmark model Gini of 0.38 reported in the lecture notes.

Part III

Figure 1d plots the consumption-equivalent welfare gain $\lambda(a, s)$ from moving from the incomplete-markets equilibrium to the complete-markets allocation. The welfare gain is positive for households with sufficiently low asset holdings and becomes negative for households with sufficiently high asset holdings. Thus, poorer households tend to prefer the complete-markets allocation, while wealthier households tend to prefer the incomplete-markets equilibrium.

Under complete markets, consumption is constant across households and states and is equal to average earnings,

$$c^{FB} = \pi_e e + \pi_u u = 0.971698.$$

The corresponding lifetime utility is

$$W^{FB} = -4.252524.$$

Under incomplete markets, average lifetime utility is

$$W^{INC} = \sum_{a,s} \mu(a, s) v(a, s) = -4.454135.$$

Thus, average welfare is higher under complete markets than under incomplete markets.

For each state (a, s) , the consumption-equivalent welfare gain is computed as

$$\lambda(a, s) = \left[\frac{W^{FB} + \frac{1}{(1-\alpha)(1-\beta)}}{v(a, s) + \frac{1}{(1-\alpha)(1-\beta)}} \right]^{\frac{1}{1-\alpha}} - 1.$$

Averaging these gains over the stationary distribution gives

$$WG = \sum_{a,s} \lambda(a, s) \mu(a, s) = 0.001358.$$

Therefore, the average household would be willing to give up approximately

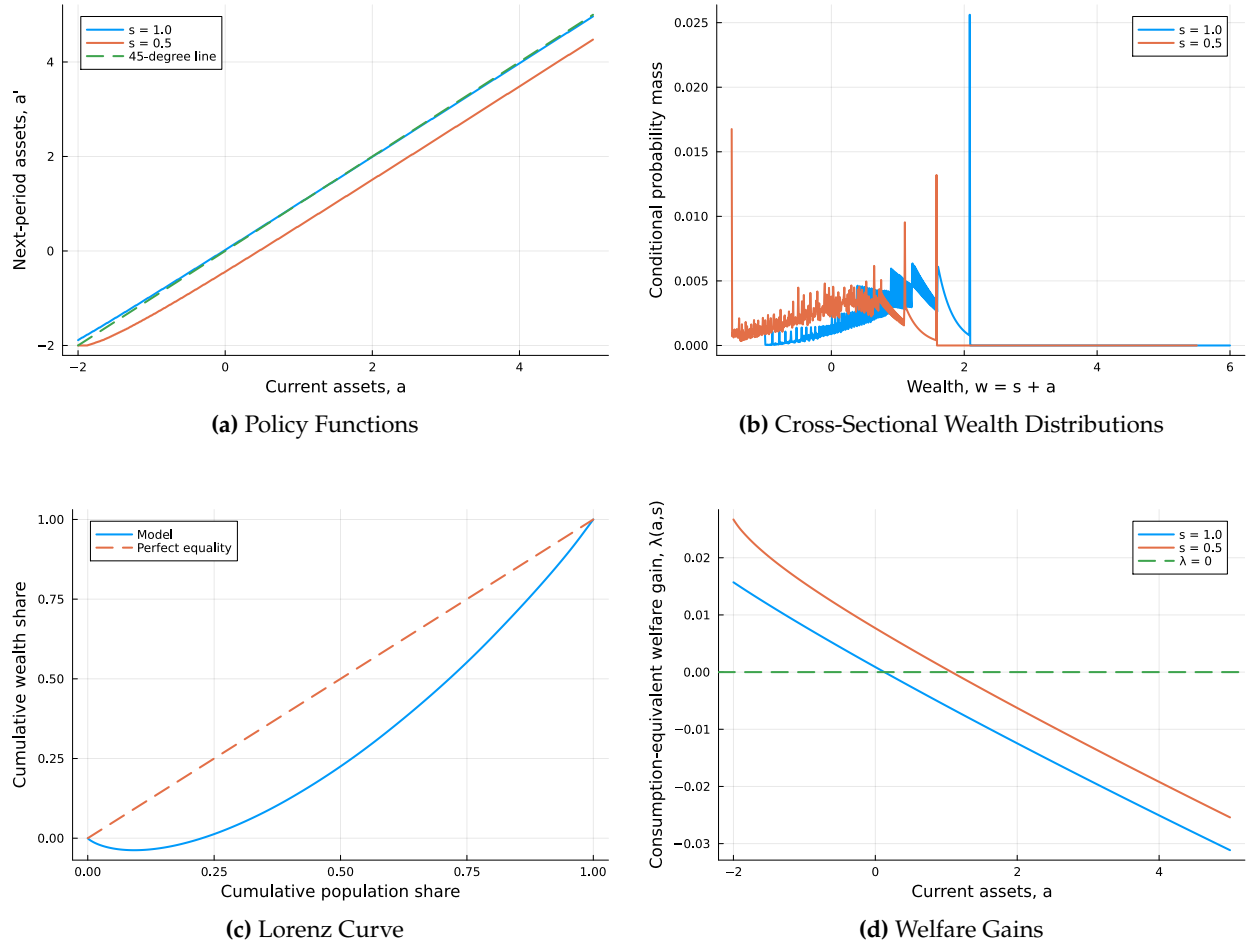
$$100 \times WG = 0.1358\%$$

of consumption in every future period in order to obtain the complete-markets allocation.

Finally, the fraction of the stationary population that weakly prefers the complete-markets allocation is

$$\sum_{a,s} \mathbf{1}\{\lambda(a, s) \geq 0\} \mu(a, s) = 0.540062.$$

Figure 1: Incomplete-Markets Equilibrium



Thus, approximately 54.01% of households would favor moving from the incomplete-markets equilibrium to complete markets.